Grade 10 Physical Sciences: Motion in one dimension

Physical Sciences · Grade 10 · 50 marks · 120 minutes

Educator marking memo

Suggested answers and marking points. Accept equivalent correct responses appropriate to the grade.

Assessment

Read the stimulus and each question carefully.

> *A trolley moves along a straight east-west track; take east as positive. During its first 6 s, it moves east from $x = -4$ m to $x = +20$ m in 4 s, then reverses and moves west to $x = +8$ m at $t = 6$ s. Assume constant velocity on each segment. In a separate later interval, its velocity changes from $+3$ m·s^{-1} to $+15$ m·s^{-1} in 4 s.*

1.1 (8 marks) Distinguish distance and displacement, then calculate both for the trolley's first 6 s of motion.

- Defines distance as total path length. (1 mark)
- Defines displacement as change in position with direction. (1 mark)
- Identifies the start at -4 m, turning position at +20 m and final position at +8 m. (1 mark)
- Calculates the eastward segment as 24 m and westward segment as 12 m. (1 mark)
- Adds segment lengths to give distance 36 m. (1 mark)
- Calculates displacement as +8 - (-4) = +12 m. (1 mark)
- States displacement is 12 m east. (1 mark)
- Uses metres for distance and displacement. (1 mark)

1.2 (9 marks) For the first 6 s, calculate the trolley's average velocity and average speed. State one difference between them.

- Uses displacement +12 m to calculate average velocity. (1 mark)
- Uses total distance 36 m to calculate average speed. (1 mark)
- Uses the total time interval 6 s for both calculations. (1 mark)
- States average velocity = displacement / time. (1 mark)
- Substitutes values correctly and obtains +2 m·s^{-1} east. (1 mark)
- States average speed = distance / time. (1 mark)
- Substitutes values correctly and obtains 6 m·s^{-1}. (1 mark)
- Uses correct SI units. (1 mark)
- Explains that velocity includes direction and uses displacement, while speed has no direction and uses distance. (1 mark)

1.3 (10 marks) Calculate acceleration during the later interval and state what its sign means.

- Identifies initial velocity +3 m·s^{-1}. (1 mark)
- Identifies final velocity +15 m·s^{-1}. (1 mark)
- Uses time interval 4 s. (1 mark)
- Uses $a = (v_f - v_i)/\Delta t$. (1 mark)

- Calculates change in velocity +12 m·s^{-1}. (1 mark)
- Divides by 4 s. (1 mark)
- Gives +3 m·s^{-2}. (1 mark)
- Uses correct SI units. (1 mark)
- States acceleration is in the positive/east direction. (1 mark)
- Connects positive acceleration to increasing positive velocity. (1 mark)

1.4 (11 marks) Assume the later acceleration stays constant at the value from Question 1.3 for a further 3 s, starting at +15 m·s^{-1}. Calculate the final velocity and signed displacement. Draw a velocity-time graph for this interval or show the trapezium area calculation, and state the direction.

- Uses the initial velocity +15 m·s^{-1}. (1 mark)
- Uses the acceleration +3 m·s^{-2}. (1 mark)
- Uses the time interval 3 s. (1 mark)
- Chooses $v_f = v_i + a\Delta t$. (1 mark)
- Substitutes the stated values correctly. (1 mark)
- Calculates the velocity change as +9 m·s^{-1}. (1 mark)
- Gives final velocity +24 m·s^{-1}. (1 mark)
- Draws or identifies the velocity-time trapezium with endpoint velocities +15 m·s^{-1} and +24 m·s^{-1} over 3 s, or states an equivalent area method. (1 mark)
- Substitutes ½(15 + 24) × 3 or an equivalent average-velocity method. (1 mark)
- Calculates signed displacement +58.5 m. (1 mark)
- Gives the displacement direction as east and uses correct units. (1 mark)

1.5 (12 marks) Compare a position-time graph and a velocity-time graph. For each graph, name both axes and give their units; state what its gradient represents and what a horizontal line means. For the velocity-time graph, state what the signed area represents.

- Names position as the vertical-axis quantity on the position-time graph. (1 mark)
- Names time as the horizontal-axis quantity on the position-time graph. (1 mark)
- Gives metres (m) for position and seconds (s) for time on the position-time graph. (1 mark)
- Names velocity as the vertical-axis quantity on the velocity-time graph. (1 mark)
- Names time as the horizontal-axis quantity on the velocity-time graph. (1 mark)
- Gives metres per second (m·s^{-1}) for velocity and seconds (s) for time on the velocity-time graph. (1 mark)
- States that the gradient of a position-time graph represents velocity. (1 mark)
- States that the gradient of a velocity-time graph represents acceleration. (1 mark)
- States that the signed area under a velocity-time graph represents displacement. (1 mark)
- Explains that a horizontal position-time line means position is constant. (1 mark)
- Explains that a horizontal velocity-time line means velocity is constant. (1 mark)
- Clearly distinguishes the quantity shown directly by each graph. (1 mark)

Marking checks

Confirm section totals equal the stated total. Apply the grade-appropriate memo style in the Grade Profile and moderate open-ended answers before use.

Source basis: DBE CAPS Physical Sciences · fet_grade_10_12. See the grade receipt for the official PDF URL and SHA-256.

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